

SYSTEM CONTEXT REPORT

ATLAS Context and Technical Reference

A self-contained handoff for people and language models encountering the project for the first time.

Evidence language
Confirmed means supported by code, a saved setup record, or a completed test. Likely means a strong inference. Assumption identifies something that still needs a physical check.

Prepared 3 August 2026

Project ATLAS School Pilot v1 - Team Touchdown

1. Answer: what ATLAS is

ATLAS in one paragraph
Confirmed: ATLAS is a wearable AI museum guide and cultural-mediation system built by Team Touchdown. A visitor wears a small wireless camera and a Shokz OpenComm2 UC headset. A Jetson Orin NX identifies the artwork being viewed, listens to a spoken question, retrieves verified museum facts, uses an LLM to phrase a short audience-appropriate answer, and speaks it through the headset. A local dashboard gives visitor and operator controls. Motorized artwork stands are part of the design, but are not yet integrated on the new Jetson.

Design goal

ATLAS is intended to make museum interpretation conversational, personalized, multilingual, and accessible. It is not designed as an unrestricted chatbot. The artwork context, approved content pack, visitor profile, grounding checks, and safety rules constrain what it should say.

Current prototype boundary

- Confirmed: Real hardware currently includes the Orin NX, XIAO wireless camera, and Shokz headset.
- Confirmed: Real services currently include Gemini, Deepgram, and Cartesia when keys and cloud switches are enabled.
- Confirmed: Local fallbacks include Whisper STT and Piper TTS.
- Confirmed: EV3 and servo movement are disabled in the current configuration.

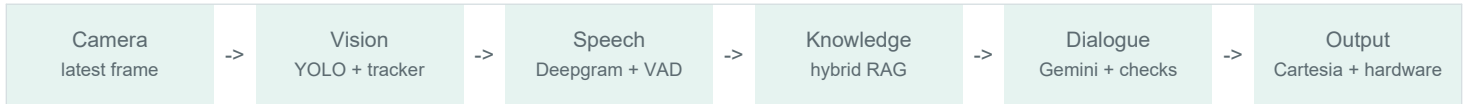
2. User experience

Actor	What they do	What ATLAS does
Visitor	Looks at an artwork and holds it near camera center.	Stabilizes detection and selects the centered artwork context.
Visitor	Hears a cue and asks a question.	Captures speech, transcribes it, retrieves verified facts, and answers aloud.
Visitor	Says 'capture this artwork' or requests manual capture.	Uses the center crop for correction when YOLO is uncertain or wrong.
Teacher/operator	Uses the local dashboard.	Starts/stops sessions, changes language/profile, types questions, overrides artwork, checks health, and manages protected controls.
Developer	Runs preflight, tests, RAG ingest/eval, and device mode.	Receives explicit component status and graceful fallbacks.

3. Physical system

Component	Role	Connection and current state
Seeed reComputer Super J401 / Orin NX 16 GB	Main edge computer	Runs camera ingest, CUDA/TensorRT, audio, RAG, dashboard, and orchestration. Tested.
XIAO ESP32-S3 Sense	Wearable eye	OV3660 camera over 2.4 GHz Wi-Fi/MJPEG at atlas-camera.local:81/stream. Tested.
Shokz OpenComm2 UC + Loop120	Visitor microphone and private audio output	USB dongle; named PulseAudio routing with ALSA fallback. Tested.
EV3 brick + three motors	Artwork stand movement	Bluetooth mailbox design; intended ports A=Starry Night, B=Mona Lisa, C=mask. Not tested on the new Jetson.
Local display	Operator view	GNOME Web can show the visitor dashboard and live YOLO camera. Tested locally.
Camera battery	Wearable power	Protected LiPo concept only; not selected or endurance-tested.

4. Software architecture



Confirmed: Components are injected through `app/dependency_container.py`. The device runtime preloads heavy resources, the session runner orchestrates one interaction, and interfaces allow mock, cloud, local, or physical adapters to be exchanged without rewriting the workflow.

Layer	Main modules	Responsibility
Application	<code>app/main.py</code> , <code>device_runtime.py</code>	Startup, preload, readiness gate, device loop, capture requests, dashboard lifecycle.
Vision	<code>vision/camera_source.py</code> , <code>yolo_detector.py</code> , <code>tracker.py</code>	Fresh-frame buffering, detection, centering, confidence, stability, last-stable context, manual override.
Speech	<code>audio/deepgram_stt.py</code> , <code>silero_vad.py</code> , <code>cartesia_tts.py</code>	Streaming STT, local endpointing, streaming PCM TTS, Shokz routing, fallbacks.
RAG	<code>rag/retriever.py</code> , <code>fusion.py</code> , <code>reranker.py</code>	Dense + keyword retrieval, reciprocal-rank fusion, metadata filters, reranking, context packing.
Dialogue	<code>dialogue/dialogue_engine.py</code> , <code>gemini_client.py</code>	Prompt construction, streaming generation, grounding, safety, multilingual response.
Pipeline	<code>pipeline/session_runner.py</code>	Coordinates detection, cue, STT, retrieval, LLM, sentence TTS, and hardware reset.
Dashboard	<code>dashboard/api.py</code> , <code>runtime_service.py</code>	Visitor/admin HTTP API, local UI, live camera, typed question fallback, protected controls.
Hardware	<code>hardware/base.py</code> , <code>ev3_hardware.py</code>	Safe command boundary, emergency-stop latch, optional EV3 messaging.
Storage	<code>storage/event_logger.py</code> , <code>sqlite_db.py</code>	Structured privacy-safe events and local indexes.

5. Startup workflow

Step	Action	Failure behavior
1	Load YAML, environment overrides, paths, and secrets.	Invalid settings are rejected; secrets are not logged.
2	Construct the dependency container for the selected run mode.	Mock or fallback adapters remain available.
3	Preload camera, TensorRT/YOLO, RAG/embedder, STT, TTS, and Gemini client.	Required device components block readiness; dashboard failure is reported separately.
4	Start the localhost dashboard on port 8765.	Device interaction may continue if dashboard startup fails.
5	Print readiness and wait for operator input before conversation begins.	This prevents ATLAS from speaking before all slow resources are ready.
6	Enter the camera-driven loop.	Fresh-frame reader discards backlog to avoid delayed detections.

6. Main interaction workflow

Stage	Detailed behavior
Artwork acquisition	The camera reader keeps only the newest frame. YOLO returns class, confidence, normalized box, and center score. The tracker requires stable frames and validates artwork IDs.
Gaze trigger	A detection must be centered and held for the configured two seconds. A latched artwork cannot retrigger until gaze clears.
Visitor cue	A short two-note cue signals that the microphone is listening.
Speech capture	Deepgram receives streaming audio; Silero detects speech start/end using pre-roll, minimum speech, and silence thresholds. Whisper is the offline fallback.
Language handling	Deepgram runs multilingual recognition. The prompt tells Gemini that speech recognition can produce plausible errors; a narrow known French homophone repair handles the observed La Joconde case.
Retrieval	The transcript, artwork ID, language, and audience level become a retrieval query. Dense and keyword candidates are fused and packed with source/chunk IDs.
Dialogue	Gemini receives the question, profile, language, and retrieved facts. Prompt injection and safety rules constrain the answer; grounding checks reject unsupported output.
Streaming answer	Complete validated sentences are emitted to TTS while Gemini continues producing later text. This hides much of the generation/playback latency.
Completion	Cartesia streams 24 kHz PCM to Shokz. Piper is the local fallback. After speech, hardware should return artwork stands to their neutral state once EV3 is enabled.

7. Vision workflow

- Input: MJPEG stream at 640x480 from the helmet camera.
- Freshness: A background reader discards queued frames; consumers receive the newest frame.
- Inference: TensorRT FP16 is preferred; PyTorch CUDA is the fallback.
- Selection: The best detection balances class confidence and distance from frame center.
- Thresholds: General confidence is 0.24; the Tutankhamun mask uses 0.45 to reduce false positives.
- Trigger: Center threshold 0.35 and hold duration 2.0 seconds.
- Current automatic classes: Mona Lisa, The Starry Night, and Tutankhamun's mask.

Manual capture workflow

When automatic detection is wrong, the operator can press c then Enter, use the dashboard Capture button, or say the translated equivalent of 'capture this artwork'. ATLAS takes only an in-memory center crop, JPEG-encodes it, asks Gemini to identify one of the approved candidates, and pins the returned artwork in the tracker. The image is not stored. A future Shokz button press is intended to call the same request hook, but the Linux button event is not yet known.

8. Speech workflow

Function	Primary	Fallback	Key behavior
STT	Deepgram Nova-3 multilingual streaming	faster-whisper small, CPU int8	Silero VAD provides local endpointing and pre-roll so first words are retained.
TTS	Cartesia Sonic 3.5 streaming	Piper local voices	Raw 24 kHz PCM streams to the selected Shokz sink; no temporary cloud audio file is required.
Cue	Generated two-note WAV	Same local playback path	Visitor hears when to begin speaking.
Routing	Named PulseAudio Shokz devices	ALSA	Avoids unstable card numbers and monitor pseudo-inputs.

9. Retrieval-augmented generation

Question normalize	->	Dense semantic	->	Keyword FTS5/BM25	->	RRF rank fusion	->	Rerank metadata + intent	->	Context tagged chunks
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- Dense search: sentence-transformers / Chroma in real modes; a dependency-free mock embedder in dev mode.
- Keyword search: SQLite FTS5 BM25 with pure-Python fallback.
- Filtering: artwork, language, educational level, student allowance, and verification status.
- Fusion: reciprocal rank fusion with k=60 avoids comparing incompatible raw scores.
- Granularity: long entries are split at sentence boundaries with a 55-word cap.
- Language: English and French content are native; Spanish and Italian may retrieve English facts while the answer remains in the visitor's language.

Current content pack

Artwork	Automatic YOLO	RAG/manual context
Mona Lisa	Yes	Yes
The Starry Night	Yes	Yes
Tutankhamun's mask	Yes	Yes
Sunflowers	No	Yes
Liberty Leading the People	No	Yes
Girl with a Pearl Earring	No	Yes
The Great Wave off Kanagawa	No	Yes

10. Dialogue, grounding, and safety

- Prompt: identifies ATLAS as a museum guide, names approved artworks, sets language/profile, requests short natural speech, and warns about STT errors.
- Cloud model: Gemini 2.5 Flash is the current prototype LLM; thinking budget is zero on the latency path.
- Grounding: response terms and cited chunk IDs are compared with retrieved context. Unsupported content is replaced with a safe fallback.
- Prompt injection: suspicious visitor instructions are rejected before the LLM.
- Output safety: a final content filter runs before any sentence is spoken.
- Hardware isolation: LLM text never directly controls motors. Hardware commands originate from reviewed application logic.

11. Dashboard workflows

Visitor dashboard

- Start/end session, select EN/FR/ES/IT, choose audience profile, enable audio description.
- See current artwork, stable/detecting state, confidence, and latest grounded answer latency.
- Ask typed questions and request manual artwork capture.
- View a live camera panel with the existing YOLO overlay; frames are in memory and not stored.

Admin dashboard

- Inspect health, active providers, privacy settings, logs, content packs, and RAG evaluation.
- Adjust selected LLM, speech, vision, and RAG settings through validated override files.
- Use emergency stop and clear it only with the configured admin token.
- Protected actions use the ATLAS_ADMIN_TOKEN environment variable and matching HTTP header.

12. Run modes and configuration

Mode	Purpose	Typical components
dev	Laptop development without hardware or keys	Mock vision/audio/LLM; real lightweight retrieval.
local	Real retrieval with mock peripherals	Chroma/FTS; mock camera/audio/LLM.
demo	Controlled demonstrations and simulations	Fixed/manual artwork; optional Gemini; mock hardware.
device	Jetson hardware runtime	Wireless camera, YOLO/TensorRT, Shokz, cloud/local speech, Gemini, optional EV3.

Confirmed: Configuration precedence is model defaults, YAML, dashboard overrides, then environment variables. Secrets remain in environment variables or a private .env. The repository YAML defaults to dev/mock for safety; device launches apply explicit runtime overrides.

13. Privacy and data flow

Data	Where processed	Stored?	Cloud exposure
Camera frames	Camera + Jetson	No	Only a manual center crop may be sent to Gemini when capture is explicitly requested.
Microphone audio	Jetson + Deepgram when enabled	No raw storage	Streaming question audio goes to Deepgram.
Transcript	Jetson	Logging off by default	Question text goes to Gemini.
Retrieved facts	Jetson RAG	Local content/indexes	Selected text chunks go to Gemini.
Answer text	Jetson + Cartesia when enabled	Only privacy-safe event metadata	Answer sentence text goes to Cartesia.
Identity	Not required	No names/face data	None by design.

14. Failure and fallback behavior

Failure	Expected behavior
Camera/network interruption	Newest-frame reader reconnects; no artwork trigger occurs until frames return.
TensorRT engine failure	Detector activates the PyTorch model fallback when possible.
Deepgram failure or missing key	FallbackSTT uses local Whisper when enabled.
Cartesia failure or missing key	FallbackTTS uses local Piper when enabled.
Gemini disabled/unavailable	Mock or safe fallback response is used according to mode; ungrounded output is not spoken.
EV3 unavailable	Hardware remains optional; digital interaction can continue. Emergency stop blocks all hardware send operations.
Dashboard unavailable	Device runtime reports the dashboard failure but can continue its core interaction loop.

15. Current tested state

What a future maintainer may safely assume

Confirmed: The Orin NX software stack boots; camera and Shokz work; YOLO/TensorRT detects the original three works; grounded Gemini answers have completed a real hardware cycle; Deepgram, Silero, and Cartesia have live component tests; the dashboard and live camera view work locally. Do not assume EV3, battery endurance, remote teacher access, or the Shokz button are complete.

Area	Status	Last meaningful verification
Jetson/CUDA stack	Working with recovery caveat	Boot, package checks, CUDA imports, YOLO load.
Wireless camera	Working	Fresh frames, ~23 FPS, live dashboard JPEG.
Shokz audio	Working	Live loopback, cue, full interaction.
YOLO/TensorRT	Working for 3 classes	Live detections, benchmark, parity test.
RAG	Working for 7 artworks	143 chunks; 9/9 retrieval evaluation.
Gemini	Working	Authorized real interaction and dashboard grounded answer.
Deepgram/Silero	Working with accent caveat	Live multilingual STT; French homophone issue observed.
Cartesia	Working	Live Shokz output; ~109-150 ms first audio.
Dashboard	Working locally	Visitor/admin pages, camera endpoint, GNOME Web.
EV3/stands	Not integrated	Disabled; no new-Jetson physical test.

16. Instructions for future LLMs and developers

- Read `ATLAS_JETSON_NX_SETUP_LOG.md` before changing Jetson packages. Do not casually run a full apt upgrade.
- Treat `ATLAS_School_Pilot_v1_Phase3_1/atlas` as the current local source and `ATLAS_BASELINES/2026-08-02_pre_streaming_speech` as the pre-upgrade baseline.
- Never print, commit, or copy API keys into reports. Keys are private environment data.
- Run preflight before device testing, then test one subsystem at a time.
- Do not claim the full system is complete until EV3, final cloud-speech interaction, complete regression suite, and endurance checks all pass together.
- When modifying latency, measure separate stages: VAD endpoint, STT final, retrieval, LLM first sentence, TTS first audio, and total response.

17. Key paths and commands

Jetson project: `~/atlas/ATLAS_School_Pilot_v1_integrated`

Jetson environment: `~/atlas/venvs/atlas-school-pilot`

Start: `./scripts/start_device.sh`

Preflight: `./scripts/preflight_device.sh --open-camera`

Tests: `python -m pytest -q`

RAG evaluation: `python scripts/evaluate_rag.py`

Dashboard: `http://127.0.0.1:8765/`
Stop service: `systemctl --user stop atlas-final.service`

18. Risks and unknowns

- Confirmed: The Jetson package state is functional but mixed; a clean vendor reflash is the safest long-term baseline.
- Confirmed: No final acceptance run combines every latest component with the physical stands.
- Likely: Network quality will remain the dominant variable for camera, Deepgram, Gemini, and Cartesia latency.
- Assumption: A protected 800-1000 mAh LiPo may approach the desired camera runtime; only current measurement and a timed test can confirm it.

19. Source register

Source	Purpose
ATLAS_JETSON_NX_SETUP_LOG.md	Jetson setup/recovery, installed versions, hardware checks, complete interaction records.
ATLAS_Hardware_Guide_Beginner.pdf	Original intended bring-up stages, device roles, and completion checkpoints.
docs/architecture.md	Core architecture and run-mode design.
docs/hardware_integration_status.md	Measured camera/vision results, RAG content, and physical status.
config/settings.yaml	Current providers, thresholds, privacy, dashboard, and disabled hardware.
src/atlas/ and tests/	Implemented workflows and regression coverage.
data/content_packs/demo_pack/manifest.json	Current artwork inventory.